LARP: Latent Astrophysical Residual Pipeline

Decoupled Physics Reduction & Population-Scale Empirical Bayes Calibration for Exoplanet Transmission Spectroscopy

PHYSICS REDUCTION CHAIN (STAGES 1–6)

Stage 1: Data Governance & Partitioning

CERTIFIED

Input: raw AIRS-CH0 + FGS1 readouts (1,100 planets)

1:1 star-planet audit | 5-fold CV + 10% frozen vault

C01_manifest, C02_cv_folds

C01/C02

Stage 2: Detector Calibration & Paired CDS

CERTIFIED

Input: 30 fps up-the-ramp telemetry, dark/flat cubes

-18.3 dB noise reduction

calibrated float64 cubes

float64 cubes

Stage 3: Curved Horne Extraction + PSF Telemetry

CERTIFIED

Input: non-linear trace modeling, optical distortion map

+18.4% SNR gain

C04_lightcurves, spatial_telemetry

C04_lightcurves

Stage 4: Decoupled AIRS Transit Geometry Lock

CERTIFIED

Input: AIRS white-light (t0, a/R*, i) vs. FGS1 photometer

FGS1 dt0 = 88 s offset detected

C05_geometry

C05_geometry

Stage 5: Physics-Informed Latent Continuum

CERTIFIED

Input: empirical stellar priors, limb darkening locking

S = 0.6705

C06_transmission_spectrum

C06_transmission

Stage 6: Vectorized FWL QR Systematics Marginalization

CERTIFIED

Input: 15-vector nuisance basis (orbit + drift + jitter)

$P_\perp = I - QQ^T$

15-vector basis | 14.16 ms

C07_marginalized_spectrum

C07_marginalized (Population Transfer)

POPULATION INFERENCE & CALIBRATION (STAGES 7–10)

Stage 7: SVD + Woodbury Empirical Bayes

CERTIFIED

Input: C07 spectra across 988 OOF planetary systems

8x8 core solve | S = 0.7215

C08_spectra, C08_population (988x567)

C08_spectra

Stage 8: Winn-Pont beta + Split-Conformal Calibration

CERTIFIED

Input: 4-stratum residual quantiles (FGS1, Blue, Mid, Red)

4-stratum calibration | 95.2% coverage

C10_calibrated_spectrum

C10_calibrated

Stage 9: 3-Family Moment-Matched Ensemble

CERTIFIED

Input: M1 (Physics GLS) + M2 (Ridge Learner) + M3 (EB SVD)

S = 0.7850 | D-GLL = +0.0636 nats

C09_blended_spectrum

C09_blended

Stage 10: Production Packaging + Quarantine Fallback

CERTIFIED

4-Tier Branching Fallback Ladder:
- Tier 0: Nominal Moment-Matched Posterior (98.1%)
- Tier 1: Mild Anomaly (gamma = 1.15)
- Tier 2: Severe / FGS1 Anomaly (Floor = 950 ppm)
- Tier 3: Pathological Outlier (Population Prior)

12/12 Pre-Submission Checks Passed

submission.csv (1100x567)

Figure 1: Architecture of the Latent Astrophysical Residual Pipeline (LARP), spanning deterministic detector physics reduction (Stages 1–6) and population-scale Empirical Bayes shrinkage and conformal calibration (Stages 7–10), certified on 1,100 planetary targets.